

Demand Sensing Intelligence

Research priorities supported by market signals and internal evidence

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Agentic AI / RAG / Knowledge graph

The decision: where should research invest next?

A research team needs to distinguish a new opportunity from an existing answer.

Research planning

Market question
External signals and internal coverage
Ranked research actions

Client question answering

Client question
Relevant internal passages
Cited answer and support assessment

Human decision

Commission research, refresh an existing note, maintain coverage, or monitor.

Five requirements, with visible evidence

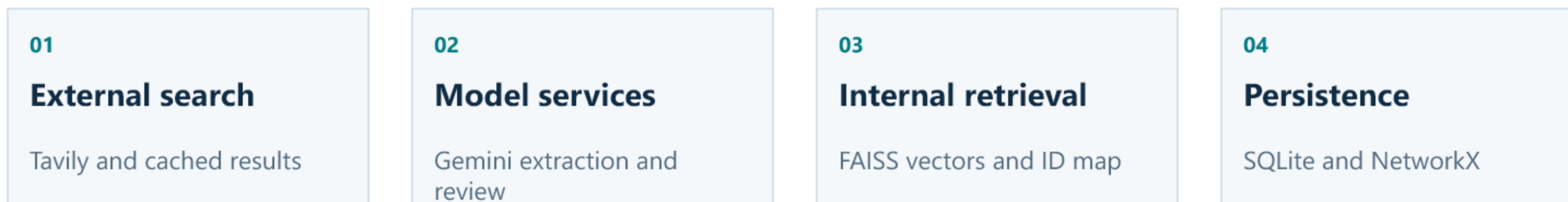
Requirement	Implementation	Evidence in the product
A. Emerging topics	News search and topic consolidation	Canonical themes with article provenance
B. Cited RAG answers	Internal retrieval and support review	Answer, document IDs and retained passages
C. Coverage gaps	Attention compared with usable coverage	Ranked gaps and proposed actions
D. Knowledge graph	Topics, entities and accepted documents	Map scoped to the selected scan
E. Agentic orchestration	Conditional LangGraph workflow	Agent trace and recorded outputs

MVP scope: sampled news attention and a simulated internal library. Historical surge detection remains future work.

Application architecture

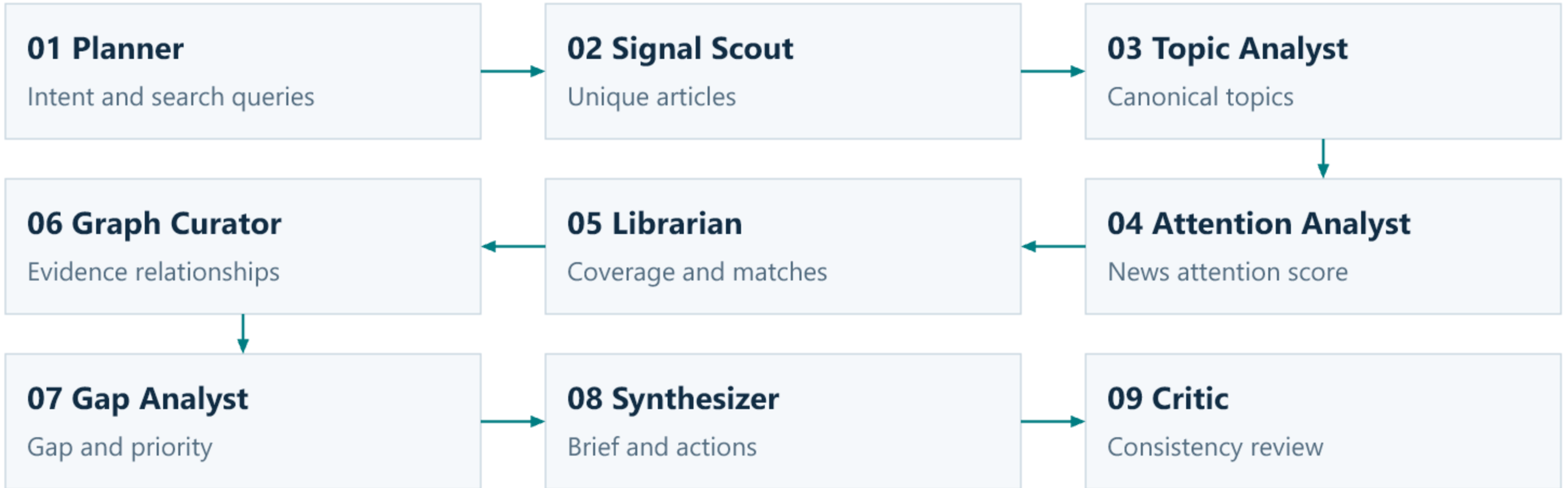


SSE streams actual agent events back to the browser



Shared state preserves the original question, articles, topics, coverage and trace across the workflow.

Nine specialists share one research state



Later nodes can read earlier fields. Articles remain in state after the system creates topic records.

Setup prepares the library before inference

SETUP

Documents become searchable

Simulated notes retain stable IDs, dates and text.
Overlapping passages become normalized embeddings.
FAISS keeps vectors and a document/chunk map.

EACH QUESTION

Queries reuse the index

Embed the question or topic.
Retrieve candidate passages.
Assess relevance or write a cited answer.

~130 words / chunk

25-word overlap

768 dimensions by default

Index preparation is separate from the market-scan workflow. Embedding represents content for retrieval.

Planner and Scout preserve intent and evidence

Query: “AI in healthcare”

Planner search queries	Scout request	Recorded run #9
AI in healthcare	Up to 8 results	8 returned
AI in healthcare emerging trends	Up to 8 results	8 returned
AI in healthcare regulation governance	Up to 8 results	8 returned

24 raw hits

23 unique articles

20 outlets

Recorded example: cached Tavily results. Output retains title, URL, domain, snippet, date and source.

Topic consolidation preserves article provenance



What the model does

Extracts themes, then names and optionally merges groups. The second call returns member group IDs.

What code preserves

Article URLs, distinct sources, dates, aliases, entities and drivers map back to each canonical topic.

Illustrative counts. One article may support multiple topics. Twelve articles do not become three articles.

Attention quantifies the observed news sample

4

Distinct outlets

5

Distinct articles

4 days

Age of newest article

Breadth 0.592

Recency 0.875

Attention = 70% × breadth + 30% × recency

0.68

Illustrative result. This measures attention in a news sample. A growth claim requires a comparable historical series.

Coverage retrieval separates similarity from relevance



Candidate selection

Exclude syndicated archive. Keep one best passage per document. Require similarity ≥ 0.62 .

Judgment context

Topic + description, document metadata and a 420-character excerpt. Full news articles are not supplied.

Similarity selects candidates. Relevance judgment determines whether a document actually addresses the topic.

An old direct match can still reveal a refresh need

Teaching document	Age	Judge verdict	Usable weight
Clinical AI Governance Playbook	500 days	Covers	0.25
Diagnostic accuracy research	30 days	Tangential	0.00
Ambient documentation research	20 days	Tangential	0.00

Coverage 0.13

Staleness 500 days

Illustrative data. Old research retains background value. Fresh but tangential documents add no coverage.

The gap score supports a specific research action

$$0.68 \times (1 - 0.13)$$

Gap $\approx$ 0.59

Recommended action: refresh

The topic has meaningful attention. Existing coverage is relevant but 500 days old.

CRITICAL PRIORITY

Update the governance playbook for analyst review.

Illustrative calculation uses unrounded inputs. Gap = attention $\times$ (1 – coverage).

The knowledge graph makes evidence inspectable

MARKET ENTITIES

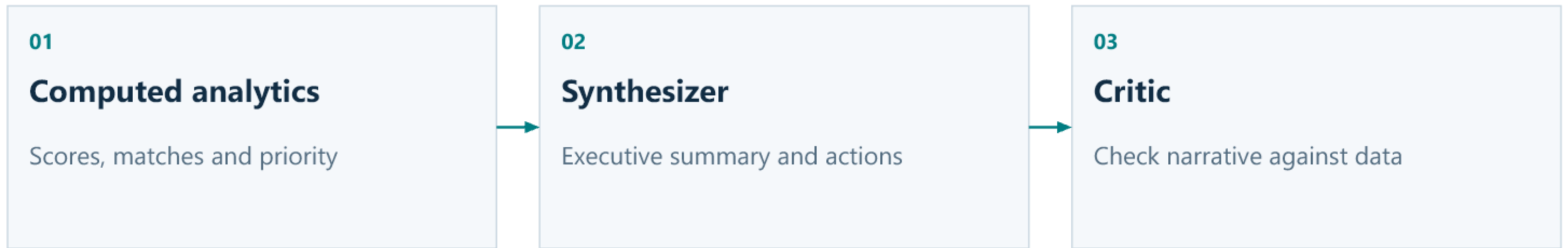
CANONICAL TOPIC

INTERNAL RESEARCH



Selected relationships from saved run #9. An accepted coverage link may represent partial or dated research.

Synthesis explains the scores and Critic checks them



Example check

The brief says coverage is 0.13. Critic compares that statement with the supplied gap table.

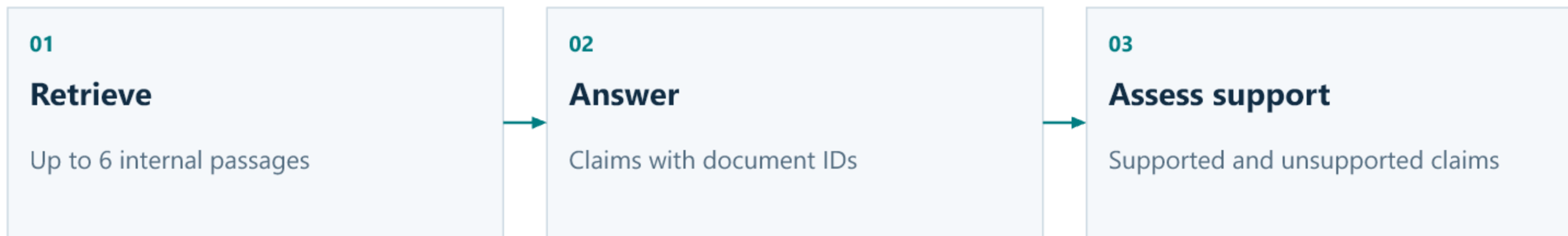
Decision boundary

The analyst reviews the recommendation. The model does not publish research or allocate a budget.

A consistency estimate evaluates the generated brief against supplied analytics. It is not external truth validation.

Library QA uses a separate retrieval path

“What governance practices does our research recommend for clinical AI?”



Coverage asks whether research addresses a topic. Library QA answers the client’s question.

No relevant passages: abstain. Returned citations must refer to documents supplied to the answer model.

Recorded runs preserve the complete demonstration

23

Unique articles

9

Canonical topics

64 s

Observed runtime

Live demonstration

Show the submitted question, agent outputs, coverage rationale and query-scoped graph.

Recorded demonstration

Open the same run from Previous runs. Its question, timestamp, evidence and graph remain together.

Observed run #9, 17 Sep 2026. Cached search results. Nine agents completed with no recorded review issues.

Quality measurement needs an analyst-labelled test set

Measure	Available today	Evaluation for a pilot
Speed and cost	Latency, provider calls and tokens	p50/p95 latency and cost per successful run
Retrieval quality	Passages and document IDs	Recall@k and citation precision
Coverage quality	Verdicts, reasons and scores	Agreement with analyst labels
Answer quality	Model support/consistency estimates	Human review of claims and omissions
Business usefulness	Proposed research actions	Accepted recommendations and analyst time saved

The recorded demo is operational evidence. It does not establish production accuracy or business impact.

Technology choices fit the prototype's scope

Choice	Why it fits	Trade-off
LangGraph	Explicit state and conditional routes	Sequential scans accumulate model latency
FAISS exact search	Simple local retrieval with a document ID map	Post-filtering can miss useful candidates
LLM relevance judge	Distinguishes same-domain text from coverage	Short excerpts and model variability limit judgment
NetworkX + SQLite	Inspectable relationships and simple persistence	Shared, multi-user scale needs further engineering
Complete run snapshots	Repeatable historical demonstrations	Evidence and traces increase storage volume

The case-study brief permits alternative technology choices. These choices support an integrated local MVP.

A staged path to a dependable research service



Pilot exit criteria

Agreed retrieval quality, useful recommendations, acceptable latency and a clear human review process.

Research planning with an inspectable evidence trail

**A market question becomes a ranked research decision.
A client question becomes a cited internal answer.**

The next step is a controlled analyst evaluation on a representative question set.

Technical appendix follows

Agent contracts: planning through attention

Agent	Reads from state	Adds or updates
01 Planner	question, forced_intent	intent, research_area, queries, plan, reasoning
02 Scout	queries, research_area, force_refresh	articles[], signal_source, scout_stats
03 Topic Analyst	articles[], research_area	topics[], topic_objects[], normalisation
04 Attention Analyst	topics and article/date/source counts	momentum and recency fields on topics

A node returns a partial state update. The shared state retains earlier fields, including the original articles.

Agent contracts: coverage through review

Agent	Reads from state	Adds or updates
05 Librarian	topic labels + descriptions, internal index	coverage by slug, matched_docs and topic scores
06 Graph Curator	topic_objects and accepted coverage	graph summary, question and node/link snapshot
07 Gap Analyst	topic attention and coverage	gaps[], priority, quadrant and portfolio
08 Synthesizer	question and computed analytics	answer, executive_summary and recommendations
09 Critic	narrative and supplied analytics	verdict, groundedness and critique_issues

topic_objects retain internal entity mappings. Serializable topics and the complete result support UI inspection and replay.

The three dictionaries preserve mapping during cleanup

Normalized phrase	surface_counter	surface_articles	surface_original
ai governance	3	[0, 1, 2]	AI Governance
agentic ai	1	[0]	Agentic AI
clinical oversight	1	[2]	Clinical Oversight

Counts and provenance have different jobs

Counter measures phrase frequency. The article map preserves evidence and enables URL deduplication.

The final topic can span several phrases

If governance and oversight merge, article 2 contributes one distinct URL to that topic.

Centroids form groups, then the model names them

1. Compare normalized vectors

score = dot(vector, centroid)
Join the nearest group at score ≥ 0.93 .
Otherwise create a new group.

2. Update the centroid

Mean of all member vectors, normalized again.
Later phrases compare with this updated centroid.

Naming-call output

label: "Clinical AI Governance"
member_groups: [0, 2]
category + description

Groups 0 and 2 identify the original phrases.
Their article indexes recover URLs, dates and entities.
The final output is a topic record with its evidence.

Coverage mathematics and missing-data rules

usable = relevance × (0.25 + 0.75 × freshness)

coverage = 0.6 × min(1, Σ usable / 5) + 0.4 × max(usable)

Input or condition	Implementation
Relevance	Covers 1.0, partial 0.5, tangential/unrelated 0
Freshness	1 through day 120, linear decline to 0 at day 450
Staleness	Age of newest accepted document
Missing date / no matches	999 is a sentinel. No matches also returns coverage 0.

Failure behavior and production boundaries

Condition	Current behavior	Remaining work
Coverage judge unavailable	Labelled partial/unrelated similarity proxy	Calibrate fallback and monitor degradation
No internal retrieval evidence	Abstain or expose unavailable coverage	Improve corpus and retrieval evaluation
Narrative generation fails	Computed summary can remain available	Retry budgets and incident monitoring
Browser stream closes	Worker can continue and save the run	Durable cancellation and resumable jobs
Live services slow or unavailable	Replay an explicitly selected DB snapshot	Operational monitoring and concurrency controls

Production work includes authentication, document access control, request-scoped telemetry and adversarial-input evaluation.